

FemCare: An XGBoost-Based Platform for PCOS Risk Assessment and Menstrual Cycle Length Prediction

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Abstract—FemCare is a women’s healthcare Platform built for menstrual cycle management and PCOS risk assessment. It runs on two XGBoost models: one classifies PCOS risk, the other predicts the length of a user’s next cycle. The PCS model pulled some wild numbers, landing at 99.85% for both accuracy and precision. Recall hit 99.70% while the F1-score clocked in around 99.77%. The ROC-UC was perfect at 1.0000 on the test set. We ran five stratified validation folds and the average accuracy came out to 99.9%. The cycle - length model is more modest. Mean absolute error is 1.89 days, RMSE is 2.29 days R^2 is 0.5274, and 57.68% of predictions land within ± 2 days of the actual cycle length. Beyond prediction, FemCare tracks periods and mood and helps users book gynecologist appointments. It’s meant to work as a decision-support tool, not a diagnostic one. An LLM-based chatbot is planned next.

Index Terms—FemCare, PCOS Prediction, Menstrual Cycle Prediction, XGBoost, Women’s Healthcare, Machine Learning, Menstrual Tracking.

1 INTRODUCTION

REPRODUCTIVE health in women is influenced by a number of factors such as menstrual patterns, symptoms, lifestyle, and hormonal changes. While most health apps only provide period logging and send out simple reminders [5], the data collected is not analysed for possible early warning signs. To go beyond this requires creating something that does more than just keep records—something that explains what the recording actually means. It is here that Artificial Intelligence and machine learning play a role. Models such as XGBoost, Random Forest, and LSTM are able to take into account a number of health factors simultaneously and detect patterns that a person would probably fail to notice themselves [3], [4]. The majority of the current tools, however, only deal with one aspect of this — either tracking, prediction, or guidance — rather than incorporating all three into one platform. FemCare is aiming to bridge that gap. It is a women’s reproductive health platform based on AI which brings together menstrual tracking, the monitoring of symptoms, an adaptive questionnaire, and machine-learning-driven risk prediction in a single system. The platform collects information on the cycle, symptoms, lifestyle factors and medical history and then passes all this data to XGBoost in order to identify reproductive health risks, particularly in the case of PCOS. The goal is to detect possible health problems as early as possible, in order that users become aware of potential risks before the situation becomes more serious. Instead of viewing the app merely

as a place for recording information, this study considers using that information to identify patterns associated with disease risk. FemCare is based on this concept – combining health tracking with AI-driven predictions to promote early awareness and enable people to take action in time. The system is not intended to take the place of a doctor or to give out a diagnosis; instead, it can assist people in spotting changes early on and in knowing when it’s appropriate to consult a healthcare professional. This method suggests that healthcare tools should be more personal, easier to understand, and actually make use of the data they collect rather than simply storing it.

1.1 Related Work & Proposed System

Menstrual monitoring and tracking apps typically record period dates and symptoms and provide reminders or simple date predictions. Work on cycle length has continued, with models that account for self-tracking artifacts and cycle variability [2], as well as wearable-based work that estimates cycle phase and the fertile window [8], [19], [22]. For PCOS, classifiers have been trained on clinical and health-record data [13], [15], [16]. These lines of work have been different: a cycle predictor does not examine PCOS, and PCOS classifiers do not necessarily use a user’s tracked history. FemCare combines a PCOS risk model, a cycle-length model, and a route to a gynecologist in one tracker.

2 PROPOSED FEMCARE ARCHITECTURE

FemCare brings together a few aspects of women’s healthcare into one platform: period tracking, cycle prediction, a PCOS screening tool, mood logging, and assistance in finding & booking appointments with gynaecologists. The

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system is divided into separate components so that user data can be gathered, processed, kept and put to use across these services.

The interface is the way users actually interact with the system. They log cycle dates, cycle details, symptoms, mood, stress, sleep and other health info the system needs. That recorded history forms the basis for the cycle's personalised predictions.

The backend is responsible for communication between the app, the database and the prediction models. It takes the input from the interface, validates it, pulls up past records if needed, formats the data for the models and sends back the results. The PCOS module is built on an XGBoost classifier. It takes the relevant health features and predicts PCOS or No PCOS, along with a prediction probability that can be shown to the user as part of the assessment.

The cycle-prediction module executes an XGBoost regression model. It considers the current cycle, previous cycle lengths, symptoms, mood, stress, sleep, and date-related. These inputs are used by the regression pipeline to figure out how long the next menstrual cycle will be. Then the expected start date of the period is determined. The cycle-prediction pipeline determines the exact feature set.

In addition to stress and sleep, the mood and wellbeing component allows users to log their mood and relevant details for the day, which also feeds into the cycle prediction model.

The app's gynaecologist appointment function lets users find gynaecologists in their area and book appointments, and serves as a bridge between the app's tracking and prediction functions and real-life professional care.

The database layer is used for storing user information, menstrual records, symptoms, mood records and other data that the system needs, with Supabase as the backend service.

We plan to extend it in the future with a chatbot based on an LLM. It's not part of the current implementation, but the plan is to take care of conversational help and general women's health questions in a future iteration.

This architecture ties together the interface, backend, database, prediction models and healthcare services into a single platform for menstrual tracking and personalised prediction.

3 METHODOLOGY

FemCare includes two different prediction modules: the first of these is a classification model (XGBoost) which predicts whether or not a woman has PCOS, and the second is a regression model (XGBoost) which predicts the length of the user's next menstrual cycle.

As for the overall structure, our method involves data preparation, preprocessing of the input data, feature selection, the development of the model, training of the model, making predictions, and evaluation. It is important that both models are trained separately: one is used to deal with the binary classification task of detecting PCOS and the other with the continuous value regression task of predicting the length of the user's menstrual cycle.

The overall process can be represented as:

Dataset Collection → Data Inspection → Data Preprocessing → Feature Preparation → Data Splitting → XGBoost

Model Training → Prediction → Performance Evaluation → Model Saving → Application Integration

In the PCOS model, pcos is used as the target variable for the prediction, while the cycle model uses Next_Cycle_Length as the target variable for the prediction.

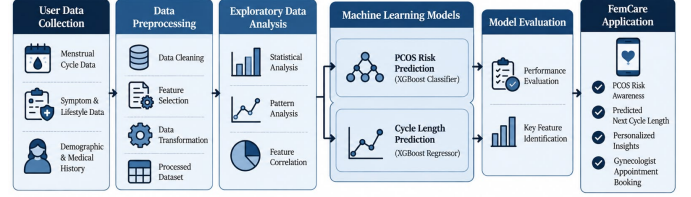


Fig. 1. Architecture of the Femcare

3.1 PCOS Prediction Methodology

3.1.1 Dataset Preparation

The PCOS prediction experiment utilizes a dataset of 10,000 records and 48 columns. The dataset was generated by the PMOS clinical literature and by the feature set of the Kerala clinical literature. The 48 columns include a range of menstrual, hormonal, physical health and other factors of a woman. This dataset was placed in the "Final_Dataset" worksheet in order to perform experiments to develop models. The target of the PCOS is the binary, with the 33,000 positive and the 6,700 negative.

At first the dataset is explored for the amount of observations, the amount of variables, the variable names, the data types and the amount of missing values. Also the distribution of the target variable pcos is explored. The values for cycle_regularity and pcos are either 0 or 2 for this experiment.

Before training the models (in the "Testing dataset" tab), the following columns are removed from the input feature matrix:

- patient_id
- blood_group_label
- cycle_regularity_label
- bmi_category
- age_group

The pcos column is used for prediction and thus has to be excluded from the input feature matrix X to use it for the target vector y. Thus, the input feature matrix

$$X = D \setminus \{\text{patient_id, blood_group_label, cycle_regularity_label, pcos, bmi_category, age_group}\} \quad (1)$$

and the target vector

y = pcos.

where D represents the original dataset.

3.1.2 Data Type Conversion

Input features are converted to numerical values before training a model. If a value cannot be converted into a number then this value is changed into a missing value. This step transforms the input matrix into a suitable matrix of numbers for training the XGBoost classifier

3.1.3 Missing-Value Treatment

A Dataset contains no missing values. A median imputer is kept in the pipeline to handle the missing input from app users at a prediction time.

3.1.4 Train-Test Data Division

After the data was processed the observations were split into two groups. 80% of the data was used to create the model. The other 20% was saved for testing. The model was built using the group. The second group was used to check how well the model worked. This helped make sure the model was accurate.

Therefore:

$$D = D_{\text{train}} \cup D_{\text{test}}$$

with approximately:

$$|D_{\text{train}}| = 80\% \quad (2)$$

$$|D_{\text{test}}| = 20\% \quad (3)$$

During partitioning a fixed random_state = 42 was used to ensure reproducibility of the experiment. The stratification was performed using the PCOS target variable to try to maintain the proportion of the two PCOS classes in the resulting subsets.

The test set used in this experiment had 2,000 cases, 1,340 of them without PCOS and 660 with PCOS.

3.1.5 XGBoost Classification

PCOS assessment was set up as a two-class prediction problem. For this the XGBClassifier model was used.

During boosting new decision trees are added to the model. Each new tree tries to fix the mistakes made by the trees before it. The process keeps going until all the mistakes are fixed. Each step helps make the model better. The model gets stronger, with each tree. The goal is to reduce the errors that were left after each step. The model becomes more accurate over time. Each tree focuses on the problems the others missed. This way the whole group of trees works together. The final result is an accurate model.

For the two-class prediction task the combined result, from the boosted trees is fed into a function. The function produces an estimate of how likely the PCOS classes.

$$P(y = 1 | X) = \frac{1}{1 + e^{-f(X)}} \quad (4)$$

where:

- X represents the input features,
- $f(X)$ represents the score that comes from the boosted trees.
- $P(y = 1 | X)$ shows the probability that is expected by the PCOS class.

TABLE 1
Hyperparameter Configuration

Parameter	Selected Value
Number of estimators	500
Maximum tree depth	5
Learning rate	0.03
Subsample	0.85
Column sample by tree	0.85
Minimum child weight	1
Gamma	0
L1 regularization	0
L2 regularization	1
Objective function	binary:logistic
Evaluation metric	logloss
Random state	42

3.1.6 XGBoost Hyperparameters

The following configuration was used for the PCOS classification model:

Hyperparameters were chosen manually and by the grid search on the validation folds.

3.1.7 Model Training

After the preprocessing steps were done and the dataset was split the XGBoost classifier was trained using the training subset.

$$Model = Train(X_{\text{train}}, y_{\text{train}})$$

After we have trained the model on the training data set, we can use the model to predict results for the test data set.

$$\hat{y} = Model(X_{\text{test}})$$

The model also generates class probabilities:

$$P(PCOS | X_{\text{test}})$$

using the predict_proba() function.

This probability will be used as the PCOS prediction probability within the FemCare application.

3.1.8 PCOS Prediction Output

The PCOS model produces two main outputs:

Class prediction

$$\hat{y} \in \{0, 1\}$$

where: 0 represents No PCOS 1 represents PCOS Prediction probability

The predicted probability that a sample belongs to the PCOS class.

The prediction function constructed here takes in user information, transforms features into numerical values and in the correct order, uses the saved imputer and then makes a PCOS class prediction and a corresponding prediction probability as output.

3.2 PCOS Model Evaluation

Here we use Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion matrix to evaluate PCOS model, and give the 5 fold cross-validation scores for the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (8)$$

The true-positive rate is:

$$TPR = \frac{TP}{TP + FN} \quad (9)$$

The false-positive rate is:

$$FPR = \frac{FP}{FP + TN} \quad (10)$$

3.3 Confusion Matrix

A confusion matrix is used to compare the actual class with the predicted class.

For the PCOS experiment, the obtained test-set confusion matrix is:

TABLE 2
Confusion Matrix Performance

Actual	Predicted	
	No PCOS	PCOS
No PCOS	1,339	1
PCOS	2	658

Therefore:

- TN=1339
- FP=1
- FN=2
- TP=658

The confusion matrix is generated using the `confusion_matrix()` function and visualized in the experiment

3.4 Five-Fold Cross-Validation

Model stability Was find out by five-fold stratified cross-validation to assess the consistency of the developed PCOS model.

For cross-validation the available data were split into five folds. In each iteration four folds were used to train the model while the remaining fold acted as the validation set. This process was repeated until every fold had been used once as a validation set.

The process can be represented as:

$$D = F_1 \cup F_2 \cup F_3 \cup F_4 \cup F_5 \quad (11)$$

For each fold:

Model_i = Train(D-F_i)

and the model is evaluated on:

F_i

The final mean accuracy is:

$$Mean\ Accuracy = \frac{1}{5} \sum_{i=1}^5 Accuracy_i$$

The standard deviation of the accuracy of the 5 folds is also summarized.

In each round, 4 folds were used for training and 1 for validation, and thus each fold was used once as a validation set. The folds were stratified and shuffled (random state 42) and the imputer was refit within each fold. We report the per-fold accuracy, its mean and the standard deviation.

3.5 Feature Importance Analysis

The ten most important features are listed in the Fig 4.

The importance is defined according to the gain importance of the XGBoost algorithm. These values reflect the contribution of the variables contained in the set to the model, but do not in any way indicate a causal relationship for a particular variable and PCOS. [10], [18]

3.6 Menstrual Cycle Prediction Methodology

The second part of FemCare, Menstrual Cycle Prediction, predicts the length of the next cycle for a woman. [2]

As opposed to the PCOS model, which predicts 2 classes (i.e. an obese or non-obese case), the cycle length model predicts a real value (i.e. the length of the next cycle).

Therefore the problem is a regression problem.

3.7 Target Construction for Cycle Prediction

The cycle dataset includes the menstrual history for many women. This history has been stored as a cycle table where information has been recorded on a per woman basis. To create a target variable for the Cycle Length prediction problem, the cycle table has been rearranged to a date ordered list of the women's menstrual history (i.e. a list of all the menstrual periods for a woman in chronological order).

- User_ID
- Period_Start_Date

This ensures that each user's menstrual records are arranged in chronological order.

The target variable for the Cycle Length prediction is defined as follows:

$$Next_Cycle_Length_t = Cycle_Length_{t+1} \quad (12)$$

In the implementation above, Cycle_Length is grouped by User_ID and then forward shifted by one period.

This means that for the cycle length of the current cycle information from this cycle is used to predict the length of the following cycle.

This means that rows for which information about future cycle lengths is not available are not included in the data sets prepared for modeling.

This includes records with missing values for current or next cycle length as well as records with current or next cycle length equal to zero or less.

3.8 Feature Engineering for Cycle Prediction

This model uses current and past cycles, symptoms and well-being, plus other information such as calendar dates.

The features are the numerals and listed in the Table 3

TABLE 3
Top 10 Features Used for Cycle Prediction

Rank	Feature
1	User_ID
2	Cycle_Number
3	Period_Start_Date
4	Period_End_Date
5	Cycle_Length
6	Period_Duration
7	Flow
8	Cramps
9	Headache
10	Bloating

3.9 Historical Cycle Features

Previous menstrual cycles give data about a persons regular cycle traits.It can also be used in the prediction process. The prediction process benefits, from this information [2].

Three previous cycle lengths are included:

$Previous_Cycle_1$

$Previous_Cycle_2$

$Previous_Cycle_3$

The average of these previous cycles is calculated as:

$$Previous_Cycle_Average = \frac{C_1 + C_2 + C_3}{3} \quad (13)$$

when all three values are available.

The standard deviation of previous cycle lengths is also computed in order to take into account the cycle length variability of the woman in the regression model.

These features allow us to utilize information from recent cycle history instead of only using information from the current cycle.

3.10 User-Level Train-Test Separation

The cycle prediction experiment uses a GroupShuffleSplit instead of simply shuffling all the records randomly.

User_ID is used as the grouping variable.

test_size is set to 20% and the random_state is set to 42. This is found in the cyclepred.py file.

The observations for a single User_ID will all be in the same group for the split, i.e. in the same training and test data.

This is particularly relevant for cycle-prediction problems, where many records may have been made by a single user. To avoid this user's data being split between the training data and the test data, the user is split at the group level.

3.11 Cycle Data Preprocessing

The numerical and categorical variables are processed separately.

Numerical preprocessing

Missing values of numerical features are imputed with the median of the respective feature.

Categorical preprocessing

Missing categorical values are filled with the most frequent value for that feature. After imputation, the categorical variables are mapped using one-hot encoding.

The encoder uses:

handle_unknown="ignore" So unseen data categories at the prediction time does not creat any issue We use a ColumnTransformer to combine the numerical and categorical transformations.

3.12 XGBoost Regression Model

The processed features are passed to an XGBRegressor.

The regression model uses the squared-error objective:

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (14)$$

where:

- y_i is the actual next-cycle length,
- $\hat{y}_i$ is the predicted next-cycle length.

This is set to minimize the difference between the model's predictions and the actual lengths of the test set's cycles.

The complete preprocessing steps and regression model have been combined into a Scikit-learn pipeline.

3.13 XGBoost Regression Parameters

The actual parameters used in your cycle prediction model are:

TABLE 4
Hyperparameter Values

Parameter	Value
Number of estimators	500
Maximum depth	5
Learning rate	0.03
Subsample	0.85
Column sample by tree	0.85
Minimum child weight	2
L1 regularization	0
L2 regularization	1
Objective	reg:squarederror
Evaluation metric	rmse
Random state	42
Number of CPU threads	-1

3.14 Cycle Prediction

After training the model above on the training data, the regression model is used on the test feature matrix to produce continuous prediction of the next cycle.

$$\hat{y} = f(X) \quad (15)$$

the predicted value is compared with the actual next-cycle length in the test set. The implementation also creates

a results table containing: actual next-cycle length, predicted next-cycle length, and prediction error.

- actual next-cycle length
- predicted next-cycle length
- prediction error

The prediction error is calculated as:

$$\text{Error} = \text{Predicted_Cycle} - \text{Actual_Cycle} \quad (16)$$

3.15 Evaluation of Cycle Prediction

We report the MAE & RMSE in the days and the R^2 on the held-out users:

The following standard regression metrics are used:

Mean Absolute Error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (17)$$

Root Mean Squared Error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (18)$$

The coefficient of determination R^2 :

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (19)$$

where $\bar{y}$ is the mean of the actual cycle lengths.

3.16 Prediction Tolerance

In addition to MAE, RMSE and R^2 , the cycle model measures the fraction of predicted cycle lengths within a given number of days of the actual cycle length.

$$\text{Accuracy}/_k = \frac{\sum_{i=1}^n I(|y_i - \hat{y}_i| \leq k)}{n} \times 100 \quad (20)$$

where $I(\cdot)$ is an indicator function that is 1 if the prediction falls within the specified tolerance and 0 otherwise.

The implementation computes:

within ± 1 day
within ± 2 days
within ± 3 days

The currently reported experiment gives:

Within ± 2 days=57.68%

3.17 Estimation of the Next Period Date

Once the predicted length of the next cycle is determined by the cycle model, FemCare can estimate the start date of the next cycle by adding the predicted cycle length to the current period's start date. The calculation is:

$$\text{Predicted Next Period} = \text{Current Period Start Date} + \text{Predicted Cycle Length} \quad (21)$$

In cyclepred.py the rounded cycle length is added to the start of the current period.

In order to make the predicted cycle length visible in the application, the number of days of the cycle length is rounded up to the nearest whole number and added to the start of the current period.

3.18 Model Saving and Application Integration

The trained PCOS model along with the imputer and feature list, and the complete prediction pipeline for the cycle is serialized for deployment. The backend loads the pickled objects and performs the same set of preprocessing steps to a new user input to make predictions.

3.19 Complete Methodology Flow

The complete methodology of FemCare is presented in the Fig. 2 :

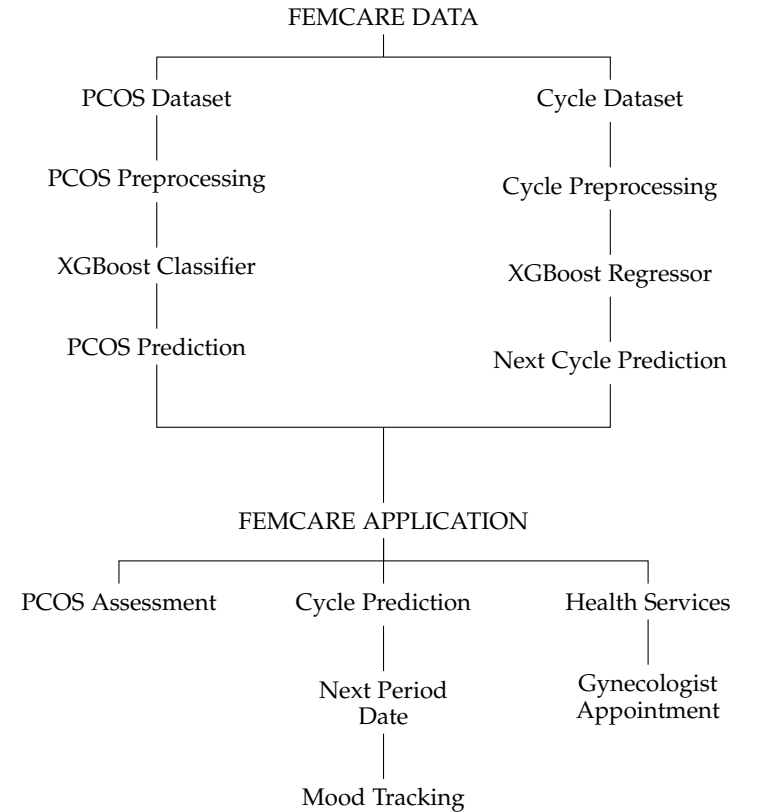


Fig. 2. System Architecture and Data Pipeline for FemCare

4 EXPERIMENT SETUP

Both models were implemented in Python in Google Colab with pandas, NumPy, scikit-learn, XGBoost. Graphics were created in Python, using Matplotlib and Seaborn. The model was stored in Joblib (the PCOS model in the XGBoost JSON format). The data sets preparation, splits, and hyperparameters are described in Section 3.

5 RESULTS OF PCOS PREDICTION

On the test set (2000 instances of which 1340 were non-PCOS and 660 PCOS), the classifier reached with the following output in table ?? & table 5

Outcomes of Cross-Validation Done in Five Different Ways The confusion matrix table 2 was used to evaluate the classification results. An ROC curve was also generated from the predicted probabilities of PCOS.

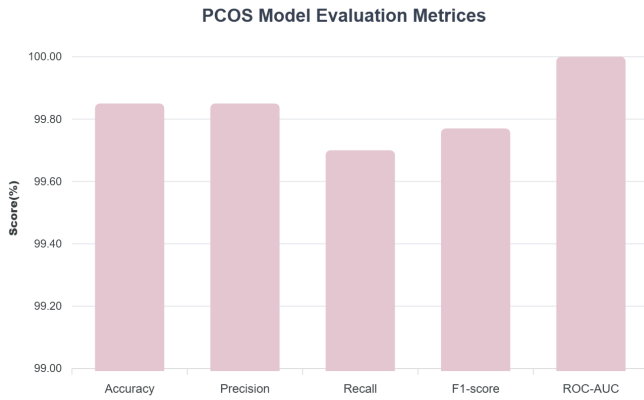


Fig. 3. Performance of the PCOS XGBoost Classification Model

TABLE 5
Measures

Measure	Result
Mean Accuracy	99.93%
Standard Deviation	0.07%

5.1 Result of Essential Features

Feature importance analysis was carried out to identify the variables that contributed most to the predictions made by the trained PCOS model. The analysis showed that follicle-related variables had high importance values.

The three most important features are follicle_no_right, total_follicles, and follicle_no_left.

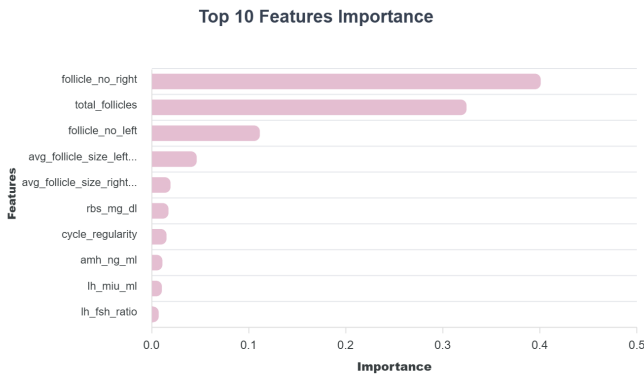


Fig. 4. Top 10 Features Importance

The findings suggest that follicle counts and follicle-size measurements contribute significantly to the model's predictions. Other contributing factors include blood glucose levels, cycle regularity, AMH, LH, and the LH/FSH ratio. The feature-importance values indicate each factor's contribution to the trained model and should not be interpreted as evidence that these factors directly cause the medical outcome. The feature-importance analysis was extracted directly from the XGBoost classifier.

5.2 Menstrual Cycle Prediction Results

The XGBoost regressor was able to achieve an MAE of 1.89 days and an RMSE of 2.29 days on the held-out users

Table 6. An R^2 of 0.5274 suggests that the model explains roughly 53% of the variance in next-cycle length on this test set. 57.68% were within ± 2 days.

5.3 Comparison of the Two Prediction Tasks

The two experiments address different prediction problems and therefore use different evaluation measures.

TABLE 6
Overall Experimental Results

Model	Task	Main Result
XGBoost Classifier	PCOS Prediction	99.85% Accuracy
XGBoost Classifier	PCOS Precision	99.85%
XGBoost Classifier	PCOS Recall	99.70%
XGBoost Classifier	PCOS F1-score	99.77%
XGBoost Classifier	ROC-AUC	1.0000
XGBoost Classifier	5-Fold CV Accuracy	99.93% $\pm$ 0.07%
XGBoost Regressor	Next Cycle Prediction	1.89 days MAE
XGBoost Regressor	Next Cycle Prediction	2.29 days RMSE
XGBoost Regressor	Next Cycle Prediction	0.5274 R^2
XGBoost Regressor	Within ± 2 days	57.68%

The PCOS classification has achieved a very good accuracy score on the used testing set, while the cycle prediction experiment has shown a moderate prediction power with a mean absolute error of 1.89 days. The different task types explain these results, since predicting a binary label (PCOS diagnosis) is generally an easier task than predicting a continuous value (cycle length).

5.4 Summary of Results

Our experimental assessment, implemented using the XGBoost classifier, sufficiently enabled differentiation between PCOS and No PCOS classes for the test data set and provided information about feature importance in predicting PCOS.

Average prediction error of the menstrual-cycle model is 1.89 days. Half of all tested cases were estimated within a margin of two days from the real cycle length. This information should be used as an estimate of menstrual cycle length, since cycle length for any individual can vary from cycle to cycle and between women.

These results are obtained from experiments performed on specific data sets. In order to consider such models for clinical use, further evaluation using independent data and so-called external validation needs to be performed.

6 LIMITATIONS

As a comprehensive system for supporting women with reproductive health issues, FemCare includes several features. In the current experimental study, however, some of these features are not examined.

6.1 Dataset and External Validation

The models within the prototype have only been evaluated using the given data and not been externally validated using a completely different, independent population. Thus, there is no guarantee for the given results being representative for other populations or for use in different health care environments.

6.2 PCOS Prediction

PCOS prediction performance reached 99.85% accuracy and ROC-AUC of 1.0000. All results must be externally validated on a different and independent dataset. Also, instead of a median imputation of missing values in the features before splitting data into training and test sets, we must use median imputation only on the training set.

6.3 Menstrual Cycle Variability

The length of the menstrual cycle for an individual can vary greatly from one time to another. It was found that the Menstrual Cycle prediction model did have some error with the predicted date of the next period with an average error of 1.89 days. The model did explain some of the variation in cycle length with an R^2 value of 0.5274.

6.4 User-Provided Data

User-provided data has the potential to be incomplete and of poor quality, affecting the performance of FemCare's functions.

6.5 Scope and Clinical Validation

These models are only for the prediction of PCOS and the next cycle length. In addition, FemCare is a research prototype only and not a clinically validated diagnostic tool. It needs to be clinically evaluated, larger data sets need to be collected and a long-term user study has to be conducted. Furthermore, an assessment of the required measures for privacy and security in health care has to be performed.

6.6 Future Components

Note that this experimental evaluation does not include the planned LLM-based chatbot and therefore it is not part of the results presented in this study. Also, the reliability of the functionality for booking appointments with a gynecologist highly depends on the external information about health-care services.

7 RESEARCH GAP

Reproductive health applications currently available mostly help users monitor menstrual cycles, symptoms and health data. The systems can store the health data collected and provide little support for users to detect possible health problems early.

Many prediction systems report only a model output, which leaves users without an explanation of what drives it [2], [5].

FemCare addresses these challenges by incorporating a comprehensive health-data collection system, a PCOS risk

assessment tool based on machine learning. Early detection of potential health risks and health understanding will be supported, whereas medical diagnosis and treatment must always be performed by qualified healthcare professionals.

8 FUTURE SCOPE

Future improvements for FemCare will focus on increasing the quality of the predictions made by the application as well as increasing the healthcare services it can provide to women.

LLM-Based Chatbot: A chatbot based on LLM can be integrated into FemCare application to support its users. This chatbot would provide conversational interface and help users asking general questions related to women health as well as explanations about data that the user can access within the application [3], [17].

Improved Cycle Prediction: More menstrual history and a user's lifestyle can be added to improve forecasting the start of the next cycle [25].

External Validation: We plan to run the models (PCOS as well as the cycle prediction models) on a larger independent dataset in order to verify the applicability of our models on a larger set of data including different types of people.

Wearable Device Integration: Tracking long-term menstrual health using data provided by wearable devices such as long-term monitoring of data like sleep, activity, heart rate and many other parameters [8], [19], [22].

Personalized Recommendations: The FemCare app can provide women with relevant menstrual health information based on past cycle information, her symptoms and predictions made by her for her next cycle.

Clinical Integration: Enable clinical interactions with users and their healthcare providers and for accessing additional gynecological healthcare services.

In summary, future work should aim to further improve the reliability of the predictions generated by FemCare.

9 CONCLUSION

FemCare is an web application that enables women to track menstrual health data, predict the next menstrual cycle, screen for PCOS and other menstrual health related illnesses, keep track of mood swings and enables booking of gynecologist appointments within the application. It incorporates two models based on XGBoost that reached 99.85% accuracy for the PCOS prediction and an average error of 1.89 days for the cycle prediction on the used datasets. FemCare is a support system for women's health management. It does not claim to fully replace a medical doctor or specialist. Future changes will aim to improve the current prediction models and add more health related features to FemCare in order to support long-term health of women's menstrual cycle.

DATA AND CODE AVAILABILITY

The dataset used for this study came from a publicly available database.

The source of the dataset [26] is cited.

The implementation code and trained model components are available at <https://github.com/dchokshi28/Femcare-Project>.

ETHICS

The study uses an anonymized dataset acquired from a publicly available database. This database does not include any personal information. The data were used for research and model training purposes only.

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